



Evaluation - Abstract

Viva Title: GREEN FUEL INTELLIGENCE SYSTEM (GFIS): A PHYSICS-INFORMED HYBRID DEEP LEARNING FRAMEWORK FOR METHANE YIELD PREDICTION IN ANAEROBIC DIGESTION.

Domain: MACHINE LEARNING

Faculty Evaluation

Non Letter Grade: EXCELLENT

Accepted by Faculty: Yes

Overall Comments: This is a phenomenal Ph.D Caliber work. Moving away from "Black Box" ML and moving to Physics-informed ML (PIML). A highly complex biological engineering problem (anaerobic digestion stability - turning waste to Methane gas) is getting solved rather than web based prediction. A sensor that uses measurable data (like temperature and pH) to predict hard-to-measure chemical states, combined with a hybrid AI model that is mathematically constrained by actual physics/chemistry rules (rather than just learning patterns). It was suggested to use real high-quality industrial dataset and the student responded with the current discussions and plans to acquire the same. Digital Twin would require a continuous, real-time data ingestion pipeline from physical IoT sensors. The work at present is a "what-if simulation engine" and a local FastAPI/Streamlit app. This is a "Simulation Dashboard," not a true Digital Twin. During the presentation, it was clearly explained how the AI is guided by Physics - Loss function is penalized for the impossibilities. A very good work for the nation too and this is appreciated.

Improvement Areas:

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